

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Support Vector Machine (SVM)

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
95.8%	95.8%	0.0s

MODEL CONFIGURATION

Model Type	Support Vector Machine (SVM)
Epochs	200
Support Vectors	24
Final Training Loss	0.09601
Data Split	80% Training / 20% Validation

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

POWER CONSUMPTION DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	6.42 W	Avg CPU Temp	45.8 °C
Peak Total Power	6.42 W	Peak CPU Temp	45.8 °C
Avg CPU+GPU Power	1.71 W	Energy Used	1.3 J (0.0004 Wh)
Peak CPU+GPU Power	1.71 W	Training Duration	0.0 s

MODEL EVALUATION RESULTS

Training Data — Accuracy: 95.83%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9490	0.9681
Recall	0.9688	0.9479
F1-Score	0.9588	0.9579
Support (n samples)	96	96

Validation Data — Accuracy: 95.83%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9583	0.9583
Recall	0.9583	0.9583
F1-Score	0.9583	0.9583
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	23	1
Actual: Low	1	23

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

MODEL DETAILS & TRAINING CURVES

SVM Summary -- 200 Epochs

Support Vectors	24
Final Training Loss	0.096008
Total Epochs	200

Feature Weights (Weight Magnitudes -- Top 15)

No	Feature Name	Weight (proportional bar)
1	tgs2600-std	0.2966
2	tgs2600-mean	0.2778
3	tgs2602-std	0.2628
4	tgs2600-max	0.2349
5	mq7-max	0.2283
6	mq6-std	0.2279
7	tgs2620-std	0.2258
8	tgs2600-median	0.2129
9	tgs2600-min	0.2113
10	tgs2600-range	0.2068
11	mq7-mean	0.1791
12	tgs2602-range	0.1605
13	mq135-range	0.1562
14	mq7-median	0.1512
15	tgs2611-std	0.1424

Accuracy Curve (Checkpoint)

Epoch	Training Accuracy (%)	Validation Accuracy (%)
20	95.83	95.83
40	95.83	95.83
60	95.83	95.83
80	95.83	95.83
100	95.83	95.83
120	95.83	95.83
140	95.83	93.75
160	95.83	95.83
180	95.83	95.83
200	95.83	95.83